

CS556 - Derivatives Homework

John Rizzo

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Question 1 (20 points): Use the limit definition of the derivative to exactly evaluate the derivative:

1. $f(x) = \sqrt{x+4}$
2. $f(x) = \frac{3}{x}$

Solution:

1. For $f(x) = \sqrt{x+4}$, the derivative using the limit definition is:

$$f'(x) = \lim_{h \rightarrow 0} \frac{\sqrt{(x+h)+4} - \sqrt{x+4}}{h}$$

Rationalizing the numerator:

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{(\sqrt{x+h+4} - \sqrt{x+4})(\sqrt{x+h+4} + \sqrt{x+4})}{h(\sqrt{x+h+4} + \sqrt{x+4})} \\ &= \lim_{h \rightarrow 0} \frac{(x+h+4) - (x+4)}{h(\sqrt{x+h+4} + \sqrt{x+4})} \\ &= \lim_{h \rightarrow 0} \frac{h}{h(\sqrt{x+h+4} + \sqrt{x+4})} \\ &= \lim_{h \rightarrow 0} \frac{1}{\sqrt{x+h+4} + \sqrt{x+4}} = \frac{1}{2\sqrt{x+4}} \end{aligned}$$

Thus, $f'(x) = \frac{1}{2\sqrt{x+4}}$.

2. For $f(x) = \frac{3}{x}$, the derivative using the limit definition is:

$$f'(x) = \lim_{h \rightarrow 0} \frac{\frac{3}{x+h} - \frac{3}{x}}{h}$$

Finding a common denominator:

$$\begin{aligned} &= \lim_{h \rightarrow 0} \frac{\frac{3x-3(x+h)}{(x+h)x}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{-3h}{(x+h)x}}{h} \end{aligned}$$

Simplifying:

$$= \lim_{h \rightarrow 0} \frac{-3}{(x+h)x} = -\frac{3}{x^2}$$

Thus, $f'(x) = -\frac{3}{x^2}$.

Question 2 (20 points) Find the derivative of the following functions:

1. $f(x) = 3x^3 - \frac{4}{x^2} = 9x^2 - 4x^{-2}$

2. $f(x) = (4 - x^2)^3 = u^3$ where $u = 4 - x^2$, so $f'(u) = 3u^2$ and $u' = -2x$ so by the chain rule $f'(x) = f'(u) \cdot u' = 3(4 - x^2)^2 \cdot (-2x) = -6x(4 - x^2)^2$

3. $f(x) = e^{\sin(x)}$ $f'(x) = e^{\sin(x)} \cdot \cos(x)$ by the chain rule.

4. $f(x) = \ln(x + 2)$ $f'(x) = \frac{1}{x+2}$

5. $f(x) = x^2 \cos(x) + x \tan(x)$

Using the product rule: $f'(x) = 2x \cos(x) - x^2 \sin(x) + \tan(x) + x \sec^2(x)$

6. $f(x) = \sqrt{3x^2 + 2}$

$$f'(x) = \frac{1}{2\sqrt{3x^2+2}} \cdot 6x = \frac{6x}{2\sqrt{3x^2+2}} = \frac{3x}{\sqrt{3x^2+2}}$$